

DENISON, IA, USA

WMO: 725477

Lat: 41.986N

Lon: 95.381W

Elev: 1276

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04917

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.6	(c) 0.1	(d) -13.3	(e) 2.8	(f) -3.2	(g) -8.6	(h) 3.6	(i) 1.2	(j) 29.5	(k) 26.8	(l) 26.8	(m) 23.6	(n) 5.3	(o) 330	(p) 0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	91.2	76.1	89.9	75.8	86.5	74.0	80.2	88.1	78.1	85.8	76.2	84.0	12.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	149.8	85.7	75.3	139.1	83.2	73.2	129.4	81.4	44.9	88.6	42.6	85.5	40.6	84.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	22.6	19.7	-15.0	96.0	6.8	2.7	-19.9	98.0	-23.9	99.5	-27.7	101.1	-32.7	103.0
			-13.7	83.1	6.2	1.6	-18.1	84.3	-21.8	85.2	-25.3	86.1	-29.8	87.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	22.7	25.4	37.6	49.8	60.6	71.3	74.8	71.7	65.4	51.6	38.3	25.7
	DBStd	20.97	12.10	12.54	12.27	10.02	8.87	6.62	6.12	5.50	8.60	10.13	10.81	11.65
	HDD50	3320	846	690	416	123	12	0	0	0	4	102	373	754
	HDD65	6512	1311	1108	852	464	195	19	6	9	99	431	800	1218
	CDD50	3211	0	1	31	117	340	638	768	673	466	153	23	1
	CDD65	926	0	0	1	7	58	207	309	217	111	16	0	0
	CDH74	8952	0	0	16	107	592	2090	3105	1846	1034	159	3	0
	CDH80	3086	0	0	2	20	180	752	1194	569	329	40	0	0
	WSAvg	8.6	8.9	9.2	10.3	11.2	9.5	8.1	6.6	6.0	7.2	8.7	9.1	8.7
	PrecAvg	29.3	0.8	0.9	1.9	2.8	4.1	4.1	3.3	3.7	3.8	2.2	1.2	1.0
(15) Precipitation	PrecMax	39.7	2.2	3.5	4.8	8.4	9.4	12.4	8.1	7.9	11.0	4.8	3.6	3.4
	PrecMin	15.1	0.0	0.0	0.2	0.7	1.3	0.8	0.4	0.3	0.7	0.0	0.0	0.1
	PrecStd	6.7	0.6	0.8	1.3	1.8	1.8	2.1	1.9	2.0	2.7	1.5	1.0	0.7
	PrecStd	6.7	0.6	0.8	1.3	1.8	1.8	2.1	1.9	2.0	2.7	1.5	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	54.4	62.6	75.3	83.6	91.2	93.4	95.2	93.1	91.1	84.5	72.6	55.3	
	0.4% MCWB	44.1	50.5	59.6	64.0	70.4	74.4	80.2	76.6	72.0	67.0	57.5	47.9	
	2% DB	46.0	54.4	69.6	77.3	84.4	90.3	91.3	89.7	88.0	78.7	64.4	50.1	
	2% MCWB	40.4	46.8	57.0	60.1	68.2	73.9	79.0	76.7	72.1	63.6	53.9	44.0	
	5% DB	41.3	47.6	62.6	72.5	81.0	87.9	90.0	86.0	83.8	72.8	59.4	45.3	
	5% MCWB	37.0	42.8	52.7	57.3	65.2	72.1	78.5	75.6	70.9	60.6	50.6	40.3	
	10% DB	37.2	42.9	55.4	67.8	76.7	84.3	86.4	82.2	80.8	68.5	54.7	40.9	
	10% MCWB	34.4	38.1	47.2	55.6	63.5	70.9	76.5	73.0	68.4	58.0	46.9	37.5	
	0.4% WB	45.6	54.7	66.5	67.9	74.3	79.8	83.5	80.9	77.5	71.4	60.9	52.2	
	0.4% MCDB	48.9	59.2	71.6	79.9	84.5	88.4	91.5	88.5	84.3	81.1	68.2	54.3	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	41.0	47.1	58.4	62.5	70.9	76.9	81.5	78.4	75.3	66.9	56.4	44.7	
	2% MCDB	45.6	53.6	64.0	72.5	81.1	85.8	89.2	86.0	82.7	74.3	63.4	47.9	
	5% WB	37.6	42.0	53.0	59.7	68.0	75.1	79.6	76.4	73.2	63.5	51.9	40.7	
	5% MCDB	41.1	47.3	61.9	69.7	76.8	84.1	87.5	84.0	80.7	69.7	58.4	44.8	
	10% WB	35.0	38.5	47.4	56.6	65.5	73.1	77.6	74.5	70.9	59.1	47.7	37.5	
	10% MCDB	37.6	42.6	57.2	65.6	74.0	81.8	84.9	81.6	77.9	66.4	54.8	40.8	
(35) Mean Daily Temperature Range	MDBR	16.0	17.0	20.2	21.4	20.7	20.4	18.9	19.2	22.5	21.3	19.4	15.8	
	5% DB	21.2	22.5	29.6	29.4	26.2	22.3	20.0	21.1	24.4	26.7	26.4	21.4	
	5% WB	16.0	15.8	18.0	15.8	12.5	10.6	11.0	12.2	12.1	14.3	16.1	16.0	
	MCDBR	19.3	20.8	26.2	23.9	21.6	19.3	17.6	18.7	20.0	21.6	23.4	19.0	
	MCWBR	15.1	15.4	17.2	15.3	12.0	11.5	11.6	12.2	11.9	14.7	16.3	15.1	
	MCWBR	15.1	15.4	17.2	15.3	12.0	11.5	11.6	12.2	11.9	14.7	16.3	15.1	
(40) Clear-Sky Solar Irradiance	taub	0.280	0.302	0.323	0.376	0.399	0.410	0.424	0.411	0.374	0.332	0.294	0.282	
	taud	2.422	2.378	2.403	2.275	2.281	2.306	2.254	2.296	2.359	2.464	2.496	2.443	
	Ebn at Noon	278	287	293	283	278	274	269	269	272	273	271	266	
	Edh at Noon	24	30	33	41	42	41	43	39	35	28	23	22	
(44) All-Sky Solar Radiation	RadAvg	613	935	1203	1525	1811	2007	2064	1757	1442	991	680	512	
	RadStd	56	62	109	124	133	166	126	121	93	121	45	43	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page